Data management II

The labs and practice sets of Day 2. You type your work in the do-file named on each page; this booklet is the printed copy to read, plan and write notes on. Core first, then Extension, then Challenge.

How the checks work. Every task in the do-file ends with a CHECK line. Run the file with Ctrl + D: a silent CHECK means your answer is correct; red text under a CHECK means look at that task again. Never edit a CHECK line. The solutions booklet explains the reasoning behind each answer; read it after you have tried.

Lab 1: Assemble Q1 from its parts

File `C:/NBS_Stata/day2/lab_D2_1.do`
inputs data/lfs_jan.dta, lfs_feb.dta, lfs_mar.dta,
outputs logs/lab_D2_1.log

The situation
The Q1 2026 LFS reaches the analysis team in pieces: one file per interview month, and separately a person file, a household file and the occupation coding file. Your section head wants proof that both routes give the same quarter, and a region x month table of interviews for the fieldwork report.

Task 1 Settings and the diary — Core

Write the five top lines of every course do-file. The diary is logs/lab_D2_1.log.

Task 2 Route A: the quarter from its three months — Core

Open lfs_jan.dta, then add lfs_feb.dta and lfs_mar.dta below it in one append line. Keep a variable called source that says which file each row came from.

Task 3 Each part holds its own month — Core

source is 0 for January, 1 for February, 2 for March, and int_month is 1, 2, 3. Write ONE assert line that is true only if every row came from the file of its own month.

Task 4 Route B: persons and their households — Core

Open lfs_person.dta. Join lfs_hh.dta by unique_hhd so that the merge stops if any person has no household, and creates no _merge.

Task 5 Add the occupation coding, then check the key — Core

Join lfs_coding.dta by unique_ind. Only employed persons are coded, so keep _merge this time (no assert, no nogenerate). Then prove there is still one row per person.

Task 6 The region x month table — Extension

The next line is written for you: it turns the person file into one row per region and month, with the number of persons in persons. (Part 5 explains this kind of command.) Reshape the result so that each region is one row and the three months are three columns.

Task 7 Does the table add up to the quarter? — Challenge

1. An empty cell here would mean no interviews: make any empty cell 0 in each of the three month columns. How many changes does Stata report? Write it: * Changes: ____ 2. Create total = the sum of the three month columns. 3. Run summarize on total.

A correct lab. the whole file runs from top to bottom with no red text, ends with ”end of do-file”, and logs/lab_D2_1.log exists.

Lab 2: The tables pack

File C:/NBS_Stata/day2/lab_D2_2.do
inputs data/lfs_teach.dta
outputs output/tables_pack.xlsx, output/tables_pack.docx,

The situation
The labour statistics section wants a first sample pack for the Q1 2026 LFS: three tables in one Excel file (one sheet each), two charts, and a Word note with a table and a chart. Every number must come from this do-file, so the pack can be rebuilt in one run.

Task 1 Settings, the diary and the data — Core

Write the five top lines of every course do-file (the diary is logs/lab_D2_2.log), then open data/lfs_teach.dta.

Task 2 Table 1: persons by region and labour force status — Core

Make the table with table, then export it to output/tables_pack.xlsx, sheet T1. Use the options sheet("T1") replace so each run starts a fresh file.

Task 3 Table 2: status within each region, one decimal — Core

Same table with row percentages and nformat(%5.1f). Export it to the same file, sheet T2. This time use sheet("T2") modify so sheet T1 is kept.

Task 4 Table 3: usual hours by region and sex — Core

Rows region, columns ilo_sex; in each cell the mean and the count of ilo_job1_how_usual. Export to sheet T3 (modify).

Task 5 Two charts — Core

Chart 1: horizontal bars of the share employed (the mean of employed) by region, with a title, ytitle("Share employed") and the source note. Export as output/lab2_fig1.png. Chart 2: a histogram of ilo_age for working-age persons only (if !missing(employed)), with the source note. Export as output/lab2_fig2.png.

Task 6 The Word note — Core

Build output/tables_pack.docx with putdocx: a title paragraph, one sentence saying the figures are unweighted sample counts, Table 1 (run the table command again just before putdocx collect), and Chart 1.

Task 7 One chart per region — Extension

Loop over the regions (levelsof, then foreach). For each region, draw the share employed by district, with ytitle("Share employed"), and export it as output/lab2_region`r'.png.

Task 8 One sheet per region — Challenge

Loop over the regions again. For each region, make a table of ilo_sex by ilo_lfs19 for that region only, and export it to output/tables_pack.xlsx in its own sheet: Region1 ... Region7.

A correct lab. the whole file runs from top to bottom with no red text, ends with "end of do-file", and the files above exist.

Practice sets

File C:/NBS_Stata/day2/practice_D2.do. One block per part of the day; each opens its own data, so the parts can be done in any order. Type your number after nbscheck where the file asks for it.

Part 1: Stacking months

P1.1 Core Append March below February. How many persons?

P1.2 Core Start again from February and append March with generate(file). Tabulate file.

P1.3 Core Write one assert: the result has 4,024 rows. Your assert is the check.

P1.4 Extension Cross-tabulate file and int_month. How many rows sit in a file that is not their own month? Type it.

P1.5 Challenge Open February again: duplicates report unique_hhd. How many persons live alone in their household?

Part 2: Joining files by their key

P2.1 Core Merge the household file onto the persons, bringing only weight dis_weight weights_dis. Then drop _merge.

P2.2 Core Join the coding file (1:1 on unique_ind) and tabulate _merge. Type the number matched.

P2.3 Core Start again from the persons joined with lfs_hh, join the district lookup (key district_code) and tabulate island by ilo_lfs19. Type the employed on La Digue.

P2.4 Extension Join the coding file again and prove with assert that every matched person is employed.

P2.5 Challenge Drop the first household of a copy of lfs_hh.dta, save it as output/hh_minus1.dta, merge it onto the persons. How many persons no longer match?

Part 3: Long and wide

P3.1 Core Make the file wide (one row per district). How many districts had no February interview?

P3.2 Core total: the persons of the quarter in each district, with egen rowtotal().

P3.3 Core Assert that the 26 totals add up to 5,888 (egen total() of total, then assert). Your assert is the check.

P3.4 Extension march: the percentage of each district's persons interviewed in March, shown with one decimal. Type the value for English River.

P3.5 Challenge Open cpi_divisions_wide.dta, turn it long (one row per month and division) and count the rows.

Part 4: Release tables

P4.1 Core Count persons by marital status (dem_mrt), showing the missing. Type the number of Single persons.

P4.2 Core Share of women and men in each age group: row %, one decimal, with table. Type the % of women aged 55+.

P4.3 Core Median usual hours by age group (tabstat). Type the median of the 55+.

P4.4 Extension Mean and count of usual hours by region and age group, mean with one decimal. Type the mean of Perseverance 55+.

P4.5 Challenge Which district has the highest share employed among working-age persons? Type that share, 3 decimals.

Part 5: From persons to groups

P5.1 Core One row per region: mean usual hours and the number of answers (keep the persons: preserve … restore). Type the mean of 7- Perseverance, one decimal.

P5.2 Core contract ilo_sex ilo_lfs19 (inside preserve … restore). How many rows?

P5.3 Core mage: the mean age of the person's region, on every row.

P5.4 Extension Rank the regions by share employed, highest = 1. Type the region code ranked 1.

P5.5 Challenge How many districts have more than 200 persons aged 15 and over (persons with a value of employed)?

Part 6: Charts

P6.1 Core Horizontal bars of the mean age by region, with a title and the source line.

P6.2 Core Histogram of usual weekly hours as counts, bars 5 hours wide. Type the number of bins Stata reports.

P6.3 Core Export the two charts as output/p6_age.png and output/p6_hours.png (graph export under each chart).

P6.4 Extension Sort the age bars from oldest to youngest region.

P6.5 Challenge Scatter of mean usual hours against share employed, one point per district, with district names.

Part 7: No more copy and paste (close Excel and Word first)

P7.1 Core Sex by labour force status as a table, exported to output/p7_sex.xlsx.

P7.2 Core The mean usual hours of women into B2 of sheet Women in output/p7_pack.xlsx. Type that mean, one decimal.

P7.3 Core A Word file output/p7_sex.docx with a title line and the same table.

P7.4 Extension Rebuild the Word file with the hours histogram of Part 6 (output/p6_hours.png) under the table.

P7.5 Challenge Share employed for men in B3 and for women in B4 of the Women sheet, computed by Stata. Type the women's share, 3 decimals.

Part 8: Write it once (run each exercise as one block)

P8.1 Core A local src holding "NBS, LFS Q1 2026 (sample)"; show it with display.

P8.2 Core One loop: tabulate, missing for ilo_sex, ilo_lfs19 and region.

P8.3 Core Loop over the regions: the mean age of each with summarize. Type the lowest regional mean age, 1 decimal.

P8.4 Extension Loop over the regions: one hours histogram per region, exported as output/p8_region1.png … p8_region7.png.

P8.5 Challenge Loop over the regions: write each region's number of employed into column B of output/p8_pack.xlsx, row = region code. Type the number in B4.